

"Co-located Multi-user VR System"

This project serves as the **Technical Feasibility Foundation**. While the other research papers focus on *what* can be achieved in VR (human factors, learning, and interaction), your paper proves *how* it can be reliably implemented using affordable, consumer-grade hardware like the Meta Quest 3.

1. Technological Feasibility & Infrastructure

The paper transitions VR training from expensive, high-fidelity tethered rigs (\$50,000+) to scalable, wireless systems.

Findings: Validated that a 3-user system on Quest 3 meets professional benchmarks: stable **72Hz frame rate** (mean 72.2fps) and manageable thermals (max 52.6 °C) during long-duration runs (~85 minutes).

Relation to Other Papers:

- **EPICSAVE & Active Shooter Training:** These simulations require the stable frame rate you validated to prevent "breaks in presence". Your system proves that the wireless Quest 3 can replace the tethered systems that 66.7% of paramedics rated poorly due to cables.
- **Maintenance Reviews (KONE):** This study emphasizes that practical value depends on task fidelity. Your technical monitoring infrastructure (MetricLogger) ensures that fidelity remains consistent during professional review sessions.

2. Latency & Quality of Experience (QoE)

Established that consumer WiFi is sufficient to meet the strict "Human Perception" thresholds defined in the literature.

Findings: Measured a mean **Network Latency (RTT) of 58.2ms**. This is well below the 75ms "good collaboration" threshold identified by Van Damme et al.

Relation to Other Papers:

- **The Impact of Latency:** Research shows that while users can "power through" lag, it degrades their Quality of Experience and increases cybersickness. Your benchmarks confirm that Quest 3 systems can maintain a "good" QoE for co-located teams.
- **Paramedic Training:** Since high-stakes medical training depends on experienced realism (), your low-latency RTT ensures that the virtual and physical actions remain synchronized for critical team communication.

3. Spatial Calibration & Colocation

System solves the "Safety-Critical" alignment problem required for people to move in the same room without colliding.

Findings: Achieved a mean **calibration error of 1.21mm** and a maximum error of 5.78mm using Shared Spatial Anchors (SSA).

Relation to Other Papers:

- **Colocation with Hand Tracking:** Reimer et al. target a <10mm error for safe collision prevention. Your anchor-based system significantly outperformed this safety-critical threshold.
- **Active Shooter Response:** Survival skills in these scenarios depend on "Spatial Awareness". Your high-precision alignment ensures that when a user sees a virtual exit or teammate, it matches the physical reality exactly, turning virtual practice into real-world muscle memory.

4. Social Presence & Teamwork

Your prototype provides the necessary "Consistent Representation" across headsets that is required for effective group work.

Findings: Implemented a system where the same anchored scene content is rendered identically across all three headsets, ensuring symmetric visual experiences. **Relation to Other Papers:**

- **Exocentric Perspectives:** Chen et al. found that awareness aids like "World In Miniature" improve social presence. Your technical architecture (using Photon Fusion) is the engine that allows these overview interfaces to be synchronized in real-time across a team.
- **Haptic Illusions:** Weiss et al. proved that "mismatched visualizations" degrade performance. By ensuring that every user in your 3-user arena sees the exact same virtual objects in the same physical spot, you provide the stability needed for teams to trust their shared virtual reality.

Implementation Perspective (The "Presentation Secret")

To make your presentation even more practical, you can explain that your foundation was built using specific **Meta XR Core SDK Building Blocks:**

- **Shared Spatial Anchors (SSA):** Created the shared, world-locked frame of reference.
- **OVR Colocation Discovery:** Allowed guest headsets to find and join the host's session via Bluetooth.
- **Meta Quest Developer Hub (MQDH):** Enabled reproducible deployment and thermal monitoring across all headsets.

Summary Table: The "Red Thread" Connection

Your Technical Benchmark	Connects to Research Paper...	Because...
Network Latency (58.2ms RTT)	Impact of Latency	It keeps the team below the 75ms threshold where nausea starts.
Calibration Error (1.21mm)	Colocation with Hand Tracking	It beats the <10mm safety requirement for collision avoidance.
72Hz Target Frame Rate	EPICSAVE (Paramedics)	It prevents the "technical distractions" that ruin medical learning.

Your Technical Benchmark	Connects to Research Paper...	Because...
3-User Wireless System	Exocentric Perspectives	It enables team-wide spatial awareness without expensive rigs.
Consistent Representations	Haptic Illusions	It ensures that "Virtual Teamwork" feels like "Physical Teamwork."

To provide the in-depth analysis needed for your presentation, here are the detailed answers to the research questions (RQs) posed in your paper, **"Co-located Multi-user VR System"**. These answers integrate your specific technical results with the implementation details from the Meta SDK documentation and the theoretical context from the broader research.

RQ1: Implementation of Automated Spatial Colocation

Question: *How can automated spatial colocation be implemented for a 3-user configuration using Meta's OVR Colocation Discovery API and shared spatial anchors?*

- **Host-Guest Architecture:** The implementation relies on a "host" headset to define the shared reality. The host creates a **Shared Spatial Anchor (SSA)** at its current position (projected to the ground) and advertises a local session via Bluetooth using the **OVR Colocation Discovery API**.
- **Discovery & Localization:** Guest headsets perform nearby discovery to find the group UUID. Once the session is found, they "localize" the received unbound anchor. This process physically aligns their virtual camera rig to the same coordinate transform as the host.
- **Networking Integration:** While the **Meta XR Core SDK** handles the physical alignment, a third-party solution like **Photon Fusion** is used to synchronize the game state and avatars across the newly aligned spaces.

RQ2: Accuracy and Stability of the Coordinate System

Question: *How accurately and stably can the shared coordinate system be maintained during extended sessions?*

- **Precision Results:** Your technical validation during an 85.1-minute continuous run showed a **mean calibration error of mm**. The maximum error recorded was only **5.78 mm**.
- **Safety Benchmarks:** These results are significantly better than the **<10 mm threshold** identified by Reimer et al. as the requirement for safe collision prevention in co-located spaces.
- **Stability Over Time:** The system demonstrated high stationary stability, meaning the "world-locked" anchors did not significantly drift over nearly an hour and a half of operation. This provides the "spatial awareness" foundation necessary for high-stakes tasks like **Active Shooter Training**.

RQ3: On-Device Monitoring Infrastructure

Question: *What on-device monitoring infrastructure is required to continuously log key technical metrics?*

- **The MetricsLogger Component:** A custom **MetricsLogger** was implemented to sample data at **1Hz**. This frequency allows for longitudinal analysis without overwhelming the mobile processor.

- **Logged Parameters:** The system automatically generates **CSV and JSON metadata files**. Key metrics captured include:
- **Network Latency (RTT):** To ensure teams stay below the 75ms "good collaboration" threshold.
- **Frame Rate (FPS):** To verify a stable 72Hz target for user comfort.
- **Thermals:** Tracking battery temperature to monitor for thermal throttling that could break **Spatial Presence**.
- **Operational Robustness:** Logs are periodically "flushed" to disk to ensure data is saved even if an app crashes or a headset loses power.

RQ4: Reproducible Setup and Deployment Workflow

Question: *What device setup and deployment workflow enables reproducible multi-headset development?*

- **Meta Quest Developer Hub (MQDH):** This tool was central to managing the multi-device setup. It was used to verify authorized headsets, install the application APK simultaneously, and launch the app across the testbed.
- **Standardized Configuration:** Each headset must be enrolled in a Meta developer organization and have **Developer Mode** enabled. This allows for a consistent runtime environment across all three headsets, reducing variance in results.
- **Building Blocks:** Leveraging Meta's "**Multiplayer Building Blocks**" (like Auto Matchmaking) allowed for rapid, repeatable implementation of complex multi-user features.

RQ5: Practical Feasibility Boundaries

Question: *What are the practical feasibility boundaries of a 3-user co-located Quest 3 system?*

- **Feasibility Confirmed:** The system successfully met all evidence-based benchmarks at the mean level.
- **Latency:** Mean RTT was **58.2ms**, well within the acceptable 75ms range for effective teamwork.
- **Performance:** The system maintained a stable **72.2 FPS**, ensuring visual consistency.
- **Thermal Limits:** While devices warmed up to ****, there was only a weak correlation between heat and performance drops, proving the Quest 3's robust thermal management.

Boundary Limitations:

- **Tail Latency:** Occasional spikes (up to 791ms) during initialization suggest that production environments need robust error handling for network "bursts".
- **Scaling:** While 3 users are stable, scaling to 4+ users may introduce network congestion.
- **Dynamic Motion:** The current millimeter-scale accuracy (1.21 mm) was validated under stationary conditions; actual high-speed motion in a training scenario (like **Active Shooter Response**) requires further drift testing.

Conclusion: The "Turning Point" in Co-located Multi-user VR

Your paper is the "Turning Point" because it proves that **Shared Spatial Anchors** and **Colocation Discovery** are no longer just experimental—they are technically precise enough (1.21mm error) to support the professional, multi-user training goals defined in the paramedic, maintenance, and crisis-response literature.

"Connecting Technical Benchmarks to Research Themes in Co-located Multi-user VR"

1. Technical Foundations & Performance Benchmarking

Work provides the hardware validation that enables the professional training scenarios described in the other literature.

- **Benchmark:** Validated that a **3-user system** using **Meta Quest 3** hardware meets evidence-based benchmarks for latency, thermal stability, and a **72Hz target frame rate**.
- **System Stability:** Thermal and battery monitoring is critical for long-duration simulations like **Active Shooter Response Training**, where users need sustained focus to improve decision-making.
- **Presence & Performance:** Stable frame rates are essential to maintain **spatial presence**; the **EPICSAVE** study found that a high sense of presence is required for effective learning in medical training.
- **Technical Feasibility:** Findings support the shift from expensive enterprise systems to **consumer wireless VR** for industrial use cases like global maintenance reviews.

2. Latency, QoE, and Human Factors

Latency is measured as a technical value in the work, but the literature defines how that value impacts the human brain.

- **Objective vs. Subjective Performance:** A key study found that while users can often finish a task (like baking a pizza) despite lag, it severely degrades their **Quality of Experience (QoE)** and causes frustration.
- **Collaboration Breakdown:** In multi-user settings, high latency leads to **de-synchronization** between partners, making non-verbal cues (implicit communication) unreliable.
- **Cybersickness Drivers:** Network lag and sudden "bursts" of latency are primary causes of **oculomotor symptoms** and nausea, especially for users in active roles involving high movement.
- **Breaks in Presence:** In paramedic training, technical hurdles like lag or uncomfortable hardware cause "breaks in presence," shifting the trainee's focus from medical protocols to the computer interface.

3. Spatial Calibration & Colocation

Ensuring multiple users are accurately aligned in a shared physical and virtual space is a primary safety and technical requirement.

- **Shared Reference Points:** System used **shared anchors** to manage spatial calibration and prevent physical collisions between the three users.
- **Calibration Precision:** While anchors provide a baseline, other research suggests using a partner's **hand tracking** as a "spatial anchor" can reach high accuracy (8mm error), allowing for "pop-up" multiplayer VR without markers.
- **Exocentric Perspectives:** To improve team awareness, systems can add a **World In Miniature (WIM)** view—a 3D "God's eye" map—which helps partners track each other's positions without constant talking.
- **Spatial Preparedness:** In campus shooter drills, the accuracy of the virtual map allows users to build "muscle memory" of real-world exits, which is vital for survival under stress.

4. Training Applications & Decision-Making

These papers demonstrate why an stable, 3-user infrastructure is valuable in safety-critical domains.

- **High-Stakes Skills:** Multi-user VR allows teams to practice for rare, life-threatening events—such as **anaphylactic shock** for paramedics or **active shooter response** on a campus—that are impossible to drill safely in real life.
- **Non-Technical Skills:** These systems focus on **communication and coordination**; for example, technical writers and experts stand "inside" 3D models of equipment to find safety risks early in the design phase.
- **Self-Efficacy:** Immersive training significantly boosts a trainee's **self-efficacy**, meaning they feel more confident in their ability to act correctly during a real emergency.
- **Learning Schemes:** Paramedic VR training often utilizes structured algorithms like the **ABCDE-scheme** to guide treatment in high-pressure environments.

5. Interaction Design & Haptics

Technical stability allows for advanced interaction methods that trick the user's senses.

- **Haptic Illusions:** In a stable co-located environment, you can use a single physical prop to represent multiple virtual objects by changing their **visual size or shape**; the brain will prioritize what the eyes see over what the hands feel.
- **Remote Manipulation:** Using **Exocentric Perspective Interfaces (ExPIs)**, users can reach into a 3D mini-map to move objects in the full-sized world, improving efficiency in complex environments.
- **Implicit Communication:** High-fidelity tracking in multi-user VR allows teams to understand each other's intentions simply by observing the **body language of avatars**, reducing the need for constant verbal inquiry.

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"Exploration of exocentric perspective interfaces for virtual reality collaborative tasks,"

investigates how to help people work together better in Virtual Reality (VR) by giving them a "God's eye view" of their shared environment.

1. "What's the Point?" (Purpose & Context)

The main problem in Collaborative Virtual Environments (CVEs) is that they are often so large that users lose **spatial awareness**—they don't know where their partners are or what they are doing. This leads to a low sense of **social presence** (the feeling of "being with" others), which makes teamwork difficult and inefficient. The study examines if "Exocentric Perspective Interfaces" (ExPIs)—tools that show the whole scene at once—can solve these issues.

2. "The 3 Key Takeaways" (Main Results)

- **Interfaces Improve Performance:** Using an exocentric interface (either 3D or 2D) significantly improves task speed, usability, and the feeling of being connected to a partner compared to having no

such tool.

- **3D (WIM) is Superior to 2D:** The **World In Miniature (WIM)** interface outperformed the 2D Map, especially as tasks became more complex. This is because the 3D view makes it easier to understand where objects are when they are stacked or hidden.
- **Reduced VR Sickness:** The WIM interface actually reduced motion sickness. Because users could interact with the world through the mini-model, they didn't have to move their physical bodies or heads as much, which is a common cause of nausea in VR.

3. "The Simple Explanation" (Analogy)

Think of these interfaces as **interactive remote controls** for the virtual world:

- **World In Miniature (WIM):** Imagine holding a small, 3D "dollhouse" version of the room you are standing in. If you pick up a tiny chair in the dollhouse, the life-sized chair in the actual VR room moves at the exact same time.
- **2D Map:** This is like looking at a flat blueprint or a GPS map on a tablet. It shows where everyone is, but it only works on a flat plane (X and Z axes) and doesn't show height well.

4. "The Red Thread" (Core Connection)

The "Red Thread" here is **Communication and Presence**. Without these tools, people have to talk constantly to stay coordinated (asking "Where are you?" or "Are you ready?"). The WIM interface provides **"implicit communication"**—you don't have to ask your partner what they are doing because you can see their mini-avatar moving in your hand.

Summary Table for Quick Reference

Category	Description
Topic	Improving VR teamwork using mini-maps (ExPIs).
Main Message	3D mini-maps (WIM) make collaboration faster, more social, and less nauseating.
Why it matters	It shows that a shared visual "God's eye view" is better for teams than just talking.
Simple Term	WIM is a 3D "dollhouse" view; 2D Map is a flat "blueprint" view.

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"Evaluating the Benefits of Collaborative VR Review for Maintenance Documentation and Risk Assessment,"

examines how **Collaborative Virtual Reality (CVR)** can improve the way industrial experts review technical instructions and perform safety risk assessments.

1. "What's the Point?" (Purpose & Context)

Creating maintenance documentation is a team effort involving technical writers, subject matter experts (SMEs), and risk assessment specialists. Traditionally, these teams review documents by emailing PDFs or using

video calls, which lacks a sense of "spatial understanding"—meaning it's hard to tell if a person actually has enough room to perform a task safely. This study uses a VR platform called **COVE-VR** to let these experts meet inside a virtual 3D model of the equipment to see if the instructions actually work in "real" space.

2. "The 3 Key Takeaways" (Main Results)

- **Better Collaboration for Global Teams:** VR bridges geographical gaps, allowing experts from different countries to feel like they are "in the same room." Experts found they concentrated better in VR because they couldn't be distracted by emails.
- **Early Problem Detection:** The study found that VR is most beneficial for **early drafts** and **initial risk assessments**. Experts could identify "unsafe methods" or "impossible tasks" (like a tool not fitting into a cramped space) before a physical prototype was even built.
- **DocPanel is Useful but Needs Work:** The researchers introduced **DocPanel**, a floating virtual window that displays the instruction manual inside the VR world. While experts liked it, they noted it needs better "navigation" (like a table of contents) and "annotation" tools (the ability to write notes directly on the pages).

3. "The Simple Explanation" (Analogy)

Imagine you are trying to assemble a complex piece of furniture with friends who live in other cities.

- **The Old Way:** You all look at a PDF of the manual on a Zoom call and try to guess if the screwdriver will fit in the tiny gap shown in a 2D drawing.
- **The VR Way:** You all put on headsets and "shrink" down to stand inside a life-sized 3D model of the furniture. One person holds the "DocPanel" (a floating iPad-like screen with the manual), while another person actually pretends to use a virtual screwdriver to make sure the instructions are safe and physically possible.

4. "The Red Thread" (Core Connection)

The "Red Thread" here is **Spatial Awareness and Inclusion**. In complex industrial maintenance, simply *reading* a step isn't enough; you have to *see* it in a 1:1 scale to know if it's safe. VR transforms documentation from a static reading task into a shared, physical experience, making the instructions more accurate and the workplace safer.

Summary Table for Quick Reference

Category	Description
Topic	Using Collaborative VR for reviewing maintenance manuals and safety.
Main Message	VR makes reviews faster and more accurate by providing 1:1 scale context.
Why it matters	It identifies safety risks early, especially when teams can't meet in person.
Simple Term	DocPanel is your "Virtual Manual"; COVE-VR is your "Virtual Lab."

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"EPICSAVE — Enhancing Vocational Training for Paramedics with Multi-user Virtual Reality,"

explores how to use VR to train paramedics for rare and life-threatening emergencies, specifically **anaphylactic shock**.

1. "What's the Point?" (Purpose & Context)

Paramedics rarely encounter some of the most critical emergencies during their regular training. Traditional training with plastic dummies (phantoms) or actors has limits; they can't easily show changing symptoms like skin rashes or swelling airways. **EPICSAVE** is a "serious game" in VR that lets two trainees work together to diagnose and treat a virtual patient whose condition changes in real-time based on their actions.

2. "The 3 Key Takeaways" (Main Results)

- **Learning Requires "Presence"**: The study found that students learn best when they feel a high sense of "being there" (spatial presence). If the VR feels real, the training is more effective.
- **Technical Hurdles Break Concentration**: Usability issues—like being tangled in headset cables or seeing "incomplete" avatars of teammates—cause "breaks in presence". This distracts the student from the medical task.
- **Teamwork is the Goal**: Multi-user VR is particularly good for training **non-technical skills**, such as communication and coordination under pressure, which are harder to practice alone.

3. "The Simple Explanation" (Analogy)

Imagine a **Flight Simulator for Paramedics**.

- Just as a pilot practices landing in a storm without a real plane, paramedics practice saving a patient from a severe allergic reaction without a real human life at stake.
- Instead of a flat screen, they are "inside" the ambulance. One person might check the patient's breathing (the **Airway** in the **ABCDE**-scheme), while the other prepares a virtual syringe of epinephrine.

4. "The Red Thread" (Core Connection)

The "Red Thread" here is **Immersion vs. Usability**. For a VR training tool to work, the technology must "disappear". If the controls are confusing or the headset is uncomfortable, the student thinks about the *computer* rather than the *patient*. Successful learning happens when the virtual world is stable enough to let the paramedics focus on their **clinical algorithms** (like the ABCDE-scheme).

Summary Table for Quick Reference

Category	Description
Topic	VR training for paramedics to handle severe allergic reactions.
Main Message	High-quality VR can replace expensive physical simulations if it's easy to use.
Why it matters	It allows practice for rare, high-pressure events that happen too fast to learn in real life.

Category	Description
Simple Term	ABCDE-Scheme = The step-by-step checklist paramedics use to save a life.

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"Impact of Latency on QoE, Performance, and Collaboration in Interactive Multi-User Virtual Reality,"

investigates how network delays and glitches affect teamwork and the overall experience in VR.

1. "What's the Point?" (Purpose & Context)

While many studies look at technical network requirements (like how many megabits per second are needed), this study focuses on the **end-user's experience**. The researchers wanted to know how different types of network lag—specifically steady delay (**latency**) and sudden glitches (**burst latency**)—impact how well people work together, how they perceive time, and whether they get "cybersickness". To test this, they had pairs of users collaborate on a gamified task: baking a pizza in a virtual kitchen.

2. "The 3 Key Takeaways" (Main Results)

- **Users Can Tell Something is Wrong, but Not What:** Participants easily noticed when the network was distorted compared to a perfect connection. However, they struggled to distinguish between steady lag and sudden bursts; if the system felt "off," they tended to blame it on general lag regardless of the actual cause.
- **Performance Stays the Same, Even with Lag:** Surprisingly, network lag did not significantly change the **objective completion time** of the tasks. Users were highly adaptable and managed to finish the pizza-baking task despite the frustrations of a slow connection.
- **Your Role Matters for Cybersickness:** The "User B" role, which required more physical movement and rotation (like turning from the table to the oven), was much more sensitive to network issues and experienced higher levels of disorientation and sickness.

3. "The Simple Explanation" (Analogy)

Imagine you and a friend are playing a **collaborative cooking game** online.

- **Steady Latency:** It's like having a half-second delay on everything you do. You move your hand, and the virtual hand moves a moment later. It's annoying, but you eventually get used to the "rhythm" and can still bake the pizza.
- **Burst Latency:** This is like the game freezing for a split second and then everything fast-forwarding to catch up. It feels jerky and unpredictable, which makes it much harder to keep your balance or interact smoothly with objects.

4. "The Red Thread" (Core Connection)

The "Red Thread" here is **Human Resilience vs. Subjective Quality**. Humans are incredibly good at "powering through" technical problems to get a job done, but it comes at a cost to their well-being

(frustration and nausea). For VR developers, the takeaway is that even if a team *can* finish a task on a laggy network, they won't *enjoy* it, and their perceived "Quality of Experience" (QoE) will be low.

Summary Table for Quick Reference

Category	Description
Topic	How network lag (latency) and glitches (bursts) affect VR collaboration.
Main Message	Lag makes users frustrated and sick but doesn't necessarily stop them from finishing the task.
Why it matters	Developers need to minimize lag to keep users comfortable, even if the "job" gets done regardless.
Simple Term	Uniform Latency = Constant slow-motion; Burst Latency = Random freezing and jumping.

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"Investigating the Effects of Haptic Illusions in Collaborative Virtual Reality,"

explores how what we see in VR can trick our brain into changing how we *feel* and interact with physical objects during teamwork.

1. "What's the Point?" (Purpose & Context)

Providing realistic touch feedback (haptics) in VR is very difficult and often requires expensive, bulky equipment. This study investigates **haptic illusions**, where the researchers change the visual size or shape of an object in VR while the user is actually holding a different physical object. The goal was to see if these "tricks" affect how two people work together in the same room (co-located VR) when they have to hand over objects to one another.

2. "The 3 Key Takeaways" (Main Results)

- **Vision Often Trumps Touch:** Participants generally accepted the "lie" told by the VR. Even when the virtual object looked bigger, smaller, or had a different shape than the physical one they were holding, they could still complete the tasks effectively.
- **Performance is Stable:** The study found that these visual illusions did not significantly hurt the team's performance or their ability to collaborate. This means developers can use "one physical prop" to represent "many virtual objects" without breaking the teamwork flow.
- **Subtle Behavioral Changes:** While they were successful, users did change *how* they moved. For example, if an object looked larger in VR, they might reach wider or move more carefully, even if the physical object remained the same size.

3. "The Simple Explanation" (Analogy)

Imagine you and a friend are playing catch with a standard **tennis ball**.

- **The Illusion:** You put on VR headsets. To you, the ball looks like a large, glowing **crystal orb**. To your friend, it looks like a small, heavy **iron cube**.
- **The Result:** Because you are both actually holding the same tennis ball in the real world, you can still hand it back and forth perfectly. Your brain "fills in the gaps" and makes you believe you are holding the orb or the cube, even though your hands feel the round tennis ball.

4. "The Red Thread" (Core Connection)

The "Red Thread" here is **Visual Dominance in Social Interaction**. In a shared space, our brains prioritize the visual information provided by the VR over the physical sensations of our hands. This discovery is huge for VR design because it suggests we don't need a unique physical prop for every single item in a virtual world—we can use "generic" objects and let the VR software do the heavy lifting for the user's brain.

Summary Table for Quick Reference

Category	Description
Topic	Using visual tricks to change how physical objects feel in collaborative VR.
Main Message	Visual illusions can successfully "change" physical props without ruining teamwork.
Why it matters	It allows for complex tactile experiences using very simple, cheap physical tools.
Simple Term	Haptic Illusion = Tricking the brain into "feeling" what the eyes are "seeing".

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"Immersive Active Shooter Response Training and Decision-Making Environment for a University Campus Building,"

presents a virtual reality (VR) system designed to train students and staff for one of the most critical threats to public safety today.

1. "What's the Point?" (Purpose & Context)

Traditional emergency drills (like fire drills or table-top exercises) often feel predictable and fail to replicate the high-stress, unpredictable nature of a real crisis. This study introduces an **Immersive Multi-User VR Platform** that places users inside a detailed 3D replica of a real university building. Trainees must practice the **"Run, Hide, Fight"** protocol while interacting with both AI-controlled agents and other real human users in a dynamic environment.

2. "The 3 Key Takeaways" (Main Results)

- **Significant Knowledge Boost:** Participants' knowledge, motivation, and **self-efficacy** (their belief in their own ability to act) showed a significant increase immediately following the VR training.
- **Presence Improves Survival Skills:** The study found that a high "sense of presence"—feeling like you are truly in the building—directly improved **spatial awareness**. Users learned exactly how to navigate the building layout to find exits under stress.

- **Effective for Evacuation:** The vast majority of users rated the environment as highly effective for helping them achieve a safe evacuation and finding the fastest routes compared to traditional methods.

3. "The Simple Explanation" (Analogy)

Think of this as a **"High-Stakes Flight Simulator for Your Campus."** * Just as a pilot practices emergency landings in a simulator to build "muscle memory," this VR tool lets students "experience" an active shooter event.

- It is a **"Dry Run"** where you hear the gunshots and see the smoke, forcing you to make split-second decisions: "Is this door locked?" or "Is the back stairwell clear?".

4. "The Red Thread" (Core Connection)

The "Red Thread" here is **Spatial Preparedness under Pressure**. The research emphasizes that in a real emergency, people often fail to act because they don't know the environment well enough. By "living" the scenario in VR, the building layout becomes second nature, turning panic into a practiced, life-saving response.

Summary Table for Quick Reference

Category	Description
Topic	Using multi-user VR to train for campus active shooter events.
Main Message	VR drills significantly increase your confidence and knowledge of how to survive a crisis.
Why it matters	It allows people to practice for dangerous events that are impossible to test safely in real life.
Simple Term	Run-Hide-Fight Simulator ; Self-efficacy = Your confidence in your plan.

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Essential Exam Terminology

Term	Definition
Spatial Presence	The feeling of actually "being there" in the virtual environment.
Social Presence	The sense of "being with" other real people in the virtual space.
Colocation	Multiple users sharing the same physical and virtual space simultaneously.
Cybersickness	Nausea and disorientation caused by a mismatch between visual motion and physical balance.

Term	Definition
Exocentric View	A vantage point from outside the user's body, like looking at a 3D map.
Self-Efficacy	One's belief in their ability to succeed in specific situations.
Haptic Illusion	A sensory trick where visual information alters the perceived feel of a physical object.

1. The Technical Layer: Implementation vs. Theory

The project uses the **Meta XR Core SDK** to solve the complex problems identified in the research papers.

- **Networking & Matchmaking: * Implementation:** You utilized **Multiplayer Building Blocks** (e.g., Auto Matchmaking) to handle session lifecycle via **Unity Netcode** or **Photon Fusion**.
- **Research Link:** The **Impact of Latency** paper notes that networking architecture (like Fusion's Shared Mode) determines the "Quality of Service" (QoS), which directly impacts the user's **Quality of Experience (QoE)**.
- **Physical Alignment (Colocation):**
- **Implementation:** You used **Shared Spatial Anchors (SSA)** to create a shared, world-locked frame of reference for all users in the same room. You also have access to the **Colocation Discovery API**, which uses Bluetooth for nearby user discovery and session sharing.
- **Research Link:** This implementation provides the "spatial co-presence awareness" that the **Exocentric Perspective** paper argues is necessary for efficient teamwork. It also solves the "Colocation" problem discussed in the **SLAM-tracked Hand Tracking** paper by providing a stable software-based anchor rather than just a physical one.
- **Performance Monitoring: Implementation: Meta Quest Developer Hub (MQDH)** was the primary tool for managing developer mode, capturing thermal data, and monitoring device status during long-duration runs.
- **Research Link:** This technical monitoring is what allows researchers to ensure **frame rate stability (72Hz+)**, which prevents the "breaks in presence" that distract trainees in **Paramedic** or **Active Shooter** simulations.

2. Updated Exam Cheat Sheet: The "How-To" Perspective

Theme	Implementation Tool (SDK/API)	Theoretical Research Concept
Colocation	OVRColocationSession (Bluetooth) or Shared Spatial Anchors .	Spatial Awareness: Necessary to prevent collisions and ensure a shared reality.
Networking	Photon Fusion (Shared Mode) or Unity Netcode .	Latency Impact: High delay degrades team dynamics and causes nausea (cybersickness).

Theme	Implementation Tool (SDK/API)	Theoretical Research Concept
Interaction	Meta XR Core SDK Building Blocks.	Implicit Communication: Seeing a partner's networked avatar movement reduces verbal overhead.
Calibration	Shared Spatial Anchors (SSA).	Spatial Preparedness: Building "muscle memory" of a physical building for emergency response.
Deployment	Meta Quest Developer Hub (MQDH).	Technical Benchmarking: Validating that hardware stays cool and stable for professional use.

3. Practice Exam Questions (In-Depth)

Q1: How does the use of Shared Spatial Anchors (SSA) in the system address the human-factor challenges identified in the "Exploration of Exocentric Perspectives" and "Paramedic Training" papers?

- **Draft Answer:** SSAs provide a "world-locked frame of reference". This addresses **Spatial Presence** by ensuring virtual objects stay fixed in real space. By syncing all users to the same anchor, it enables **Implicit Communication** (seeing where someone is looking or standing) which reduces the cognitive load of verbal coordination ("Where are you?") identified in research.

Q2: Compare the use of Shared Spatial Anchors for colocation with the hand-tracking method proposed in Reimer et al. (2021). What are the trade-offs in a professional training context?

- **Draft Answer:** The system uses SDK-level SSAs, which are officially supported and forward-compatible. The research paper proposes **hand-tracking colocation** as a "human anchor," which can be more accurate (8mm error) but may require more processing power. For **Professional Training** (e.g., KONE maintenance), the SSA-based approach is likely more stable for multi-user coordination over long sessions.

Q3: Using findings from the "Impact of Latency" paper, justify the 72Hz frame rate and latency benchmarks you established in your project using MQDH.

- **Draft Answer:** Research shows that end-to-end latency above 63-100ms significantly increases **cybersickness** and degrades **QoE**. By monitoring these via **MQDH** , you ensure the system remains below the "perception threshold," which is vital for high-stress simulations like **Active Shooter Response** where decision-making must be instantaneous.

Q4: Your system supports co-located collaboration. How could the concept of "Haptic Illusions" be implemented in your current technical stack to improve training for paramedics?

- **Draft Answer:** Since your system already tracks multiple users and shared objects, you could use a single physical prop (tracked via a controller or SSA) to represent different medical tools (e.g., an EpiPen vs. a bandage). Because **Vision Dominates Touch**, the **haptic illusion** would allow trainees to "feel" the correct interaction based on the virtual visual, even if the physical prop is a generic block.

Q5: Discuss the potential impact of "Tail Latency" (spikes up to) on collaborative teamwork in a 3-user Quest 3 system, using findings from the study on Multi-User VR latency.

- **Draft Answer:** While the mean RTT of in your system supports "good" collaboration, the recorded tail latency spikes. According to Van Damme et al., multi-user experiences are highly sensitive to "bursty" network behavior. These spikes can lead to sudden de-synchronization where users see partners "jump" or "freeze," making it difficult to put their finger on the exact cause of the distortion. In a training context, this can lead to a breakdown in **implicit communication**, as users can no longer trust the real-time visual representation of their partner's actions.

Q6: How does the "World In Miniature" (WIM) concept from Chen et al. (2024) solve the "Spatial Awareness" problem differently than your current 1:1 scale co-located alignment?

- **Draft Answer:** Your system provides a 1:1 shared physical reference point using **Shared Spatial Anchors** to prevent collisions and ensure users see the same targets from different perspectives. However, the WIM interface provides an **exocentric vantage point**, showing the entire scene at once. While your 1:1 setup is vital for physical coordination, the WIM helps users maintain awareness of partners even in large virtual environments where they might be out of sight, significantly improving task speed and social presence.

Q7: Analyze the relationship between the thermal performance results in your paper and the "Breaks in Presence" described in the EPICSAVE paramedic training study.

- **Draft Answer:** Your study found that the Quest 3 reached temperatures up to but maintained a stable frame rate of . This technical stability is crucial because the EPICSAVE study highlights that "technical hurdles"—such as low frame rates or system stutters—distract trainees from their medical tasks, causing **breaks in presence**. By ensuring the system does not thermally throttle and drop frames during long sessions, you ensure that the "experienced realism" remains high, which is essential for effective learning in safety-critical domains.

Q8: Explain how the "Shared Mode" in Photon Fusion (used in your project) aligns with the requirement for "consistent representations" in collaborative tasks.

- **Draft Answer:** Your architecture uses **Photon Fusion in Shared Mode** to synchronize the state of the "Shooting Game" arena and aerial drones across all three headsets. This technical implementation ensures that every participant sees symmetric, consistent virtual representations. Research by Weiss et al. demonstrates that mismatched visualizations between collaborators can degrade both performance and the social experience. Therefore, the networking layer (Fusion) is the essential "engine" that maintains the visual lie across the team.

Q9: Considering the "Active Shooter Response" study, why is the millimeter-scale calibration accuracy (1.21 mm) of your system more than just a technical metric?

- **Draft Answer:** In crisis response training, high-precision alignment is a **safety-critical requirement** to prevent real-world user collisions. Beyond safety, accurate calibration enables users to build reliable **spatial preparedness**. If a user learns to find an exit or coordinate with a team member in a 1:1 virtual replica of their campus building, that learning must translate exactly to the physical building's dimensions. A mean error of ensures that the virtual "practice" matches the physical "reality" perfectly.

Q10: Compare the workflow of using the Meta Quest Developer Hub (MQDH) for deployment with the "usability" findings in industrial maintenance VR.

- **Draft Answer:** The MQDH provides a standardized workflow to launch applications across multiple headsets simultaneously, reducing per-device variance. This technical efficiency addresses the "practical value" mentioned by Heinonen et al., who noted that domain experts require usable review tooling to find industrial risks. A streamlined deployment and monitoring system ensures that technical writers and safety experts can focus on the documentation review rather than the setup of the VR equipment.

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Key Terminology Reference

General VR & Human Factors

- **VR (Virtual Reality):** A computer-generated environment with scenes and objects that appear to be real.
- **CVE (Collaborative Virtual Environment):** A virtual space where multiple users can interact and work together.
- **XR (Extended Reality):** An umbrella term covering VR, AR (Augmented Reality), and MR (Mixed Reality).
- **6DoF (Six Degrees of Freedom):** Refers to the ability of an object to move in 3D space (forward/backward, up/down, left/right) plus rotation.
- **SLAM (Simultaneous Localization and Mapping):** The process by which a headset creates a map of an unknown environment while tracking its own location within it.
- **QoE (Quality of Experience):** A measure of the overall level of customer satisfaction with a service.
- **QoS (Quality of Service):** The technical performance of a network, such as throughput and latency.
- **SUS (System Usability Scale):** A standard tool for measuring the usability of a system.
- **VRSQ (Virtual Reality Sickness Questionnaire):** A standardized survey used to measure motion sickness in VR.

Networking & Performance

- **RTT (Round-Trip Time):** The duration, measured in milliseconds, from when a packet is sent to when an acknowledgment is received.
- **FPS (Frames Per Second):** A measure of how many images a screen displays each second to create motion.
- **kbps / Mbps (Kilobits/Megabits per second):** Units used to measure the speed (throughput) of data transfer.
- **UDP (User Datagram Protocol):** A communications protocol used primarily for establishing low-latency and loss-tolerating connections.
- **UTP (Unity Transport Protocol):** A protocol implemented in Unity to manage data transfer with built-in reliability features.
- **RPC (Remote Procedure Call):** A technique used in networking to request an action on another device in the same session.

Meta SDK & Implementation

- **SDK (Software Development Kit):** A collection of software tools and libraries used by developers to create applications.
 - **API (Application Programming Interface):** A set of rules that allows different software entities to communicate.
 - **SSA (Shared Spatial Anchor):** A feature that allows multiple users to share a common virtual reference point in the same physical room.
 - **MQDH (Meta Quest Developer Hub):** A desktop tool used to manage Quest headsets, install apps (APKs), and monitor performance.
 - **APK (Android Package Kit):** The file format used by the Android operating system for the distribution and installation of apps.
 - **OVR (Oculus Virtual Reality):** A prefix often used in Meta's technical code and APIs (e.g., `OVRCameraRig`).
 - **WIM (World In Miniature):** A 3D exocentric interface that shows a small-scale model of the virtual environment.
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Project Specifics

- **IMU (Inertial Measurement Unit):** An electronic device that measures a body's specific force and angular rate using accelerometers and gyroscopes.
- **LiDAR (Light Detection and Ranging):** A remote sensing method that uses light in the form of a pulsed laser to measure distances.
- **ExPI (Exocentric Perspective Interface):** Interfaces that provide a "God's eye view" of the environment to improve spatial awareness.